Exam 1 Topics

Chapter 1

- Definition of empiricism
- Quantitative vs. qualitative methods
- Theories vs. hypotheses
- The theory-data cycle
- Merton's scientific norms
- Basic vs. applied problems

Chapter 2

- What does it mean for research to probabilistic?
- Problems with personal experience and intuition as sources of information
- Confounds
- Biased thinking
    - Availability Heuristic
    - Confirmation bias
    - Bias blind spots
- Sources of information
    - Types of journal articles
- Platforms for finding legitimate scientific sources
- Identifying disinformation

Chapter 3

- Variables vs. constants
- Independent vs dependent variables
- Experimental control
- Operationalization/operational definitions
- Types of claims
    - Frequency claims
    - Association claims
    - Causal claims
- Types of associations (positive vs. negative vs. zero)
- Validity
    - Construct validity
    - Statistical validity
    - Internal validity
    - External validity

- The link between experiments and causal claims